



**SOUTHERN LEYTE  
STATE UNIVERSITY**

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# Book Rental System Documentation

SUBJECT: IT309 System Analysis and Design

**Submitted by:**

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**Submitted to:**

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Instructor





## Project Plan: Book Rental System Mobile App

### 1. Project Overview

**Project Name:** Book Rental System

**Project Type:** Mobile Application (Android Studio/Flutter)

**Project Duration:** 10–12 weeks (tentative)

**Project Description:**

Develop a mobile application that allows users to browse, rent, and manage books through an Android app. The app will include user registration, login, book listings, book details, rental checkout, and order history features. Admin users can manage books and rental transactions.

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### 2. Project Objectives

1. Create an intuitive, responsive mobile app for renting books.
  2. Enable users to browse and search books easily.
  3. Facilitate seamless rental checkout and history tracking.
  4. Provide admin functionality to manage book inventory and users.
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### 3. Scope

#### In-Scope

- User registration and login system
- Book listing with search and filtering by category/genre
- Book detail view
- Order/rental flow (Order Now → Success → Order List)
- Admin panel for managing books and users

#### Out-of-Scope

- Data persistence with database (Firebase)
  - Notifications or confirmation messages on successful rentals
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### 4. Deliverables

1. Android Studio Project (Flutter)
2. UI Interfaces



### 3. User and admin functionability

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## 5. Project Milestones & Timeline

| Milestone                            | Description                                              | Duration | Deadline |
|--------------------------------------|----------------------------------------------------------|----------|----------|
| Project Planning                     | Define scope, requirements, and resources                | 1 week   | Week 1   |
| UI/UX Design                         | Design screens, wireframes, mockups                      | 1 week   | Week 2   |
| Backend Setup                        | Create database schema & REST API endpoints              | 2 weeks  | Week 4   |
| Frontend Development – Core Features | Implement user login, registration, book listing, search | 2 weeks  | Week 6   |
| Frontend Development – Rental Flow   | Implement order process and order history                | 2 weeks  | Week 8   |
| Admin Module                         | Implement book management and user management            | 1 week   | Week 9   |
| Testing & QA                         | Unit testing, integration testing, bug fixes             | 1 week   | Week 10  |
| Deployment & Launch                  | Publish APK, final documentation                         | 1 week   | Week 11  |

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## 6. Technology Stack

- **Frontend:** Flutter
- **Programming Language:** Dart
- **Testing:** Mobile Phone for Flutter

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## 8. Risk Management

| Risk                   | Impact | Mitigation                                                         |
|------------------------|--------|--------------------------------------------------------------------|
| Backend delays         | High   | Set clear API milestones and parallel frontend work with mock data |
| UI/UX usability issues | Medium | Conduct early user testing and feedback sessions                   |



| Risk                         | Impact | Mitigation                                              |
|------------------------------|--------|---------------------------------------------------------|
| Database performance issues  | Medium | Optimize queries and index frequently accessed data     |
| Feature creep                | High   | Strictly follow scope and prioritize essential features |
| Android device compatibility | Medium | Test on multiple devices/emulators                      |

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## 9. Quality Assurance

- **Unit Testing:** Test individual components
- **Integration Testing:** Ensure function ability across user and admin

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## 10. Documentation & Reporting

- **Final Project Report:** Summarize deliverables, challenges, and lessons learned

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## 11. Success Criteria

- Users can successfully register, login, browse, rent, and view order history
- Admin can manage books and users efficiently
- App is stable, responsive, and bug-free

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## User Flow Wireframes

### 1. Splash Screen

- App logo and tagline

### 2. User Registration Screen

- Input fields: Name, Email, Password, Confirm Password
- Register button
- Link to Login screen

### 3. Login Screen

- Input fields: Email, Password



- Login button
- Link to Registration screen

#### 4. Home / Main Screen

- Search bar
- Book categories / genres
- Featured books carousel
- List of books with thumbnail, title, and author

#### 5. Book Listing Screen

- Filter by category/genre
- Sort by popularity, newest, or rating
- Book cards with title, author, and thumbnail

#### 6. Book Detail Screen

- Book cover image
- Title, author, description, genre, availability
- "Order Now" button

#### 7. Order Success Screen

- Confirmation message
- Order details (Book title, rental duration, order ID)
- Button to go back to Home or View Orders

#### 8. Order List / History Screen

- List of rented books
- Status (e.g., Pending, Completed)
- Option to view details

#### 9. Profile Screen

- User info: Name, Email
- Logout button

### Admin Flow Wireframes

#### 1. Admin Login Screen



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- Email and password fields

- Login button

## 2. Admin Dashboard Screen

- Overview of books, users, and orders
- Quick stats (total books, active rentals, pending orders)

## 3. Book Management Screen

- List of books with edit/delete options
- Add new book button
- Book details (title, author, genre, stock)

## 4. Order Management Screen

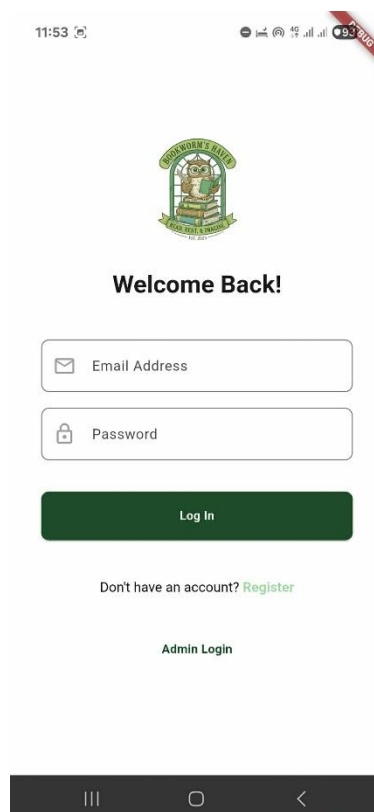
- List of current orders
- Status update options (Approved, Completed, Cancelled)

## User Interface

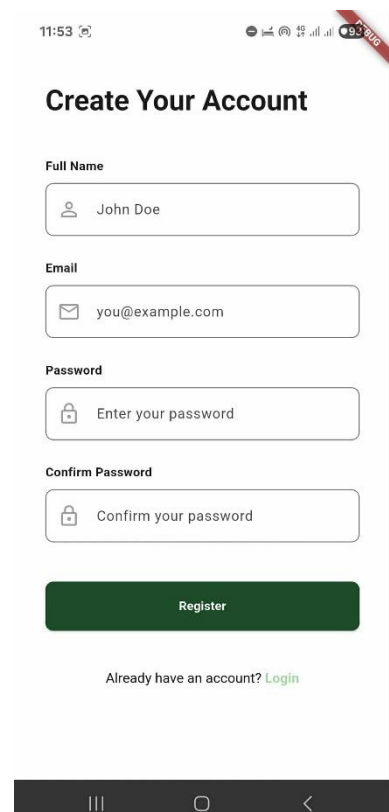
### SPLASH SCREEN



### LOGIN SCREEN



### CREATE ACCOUNT SCREEN





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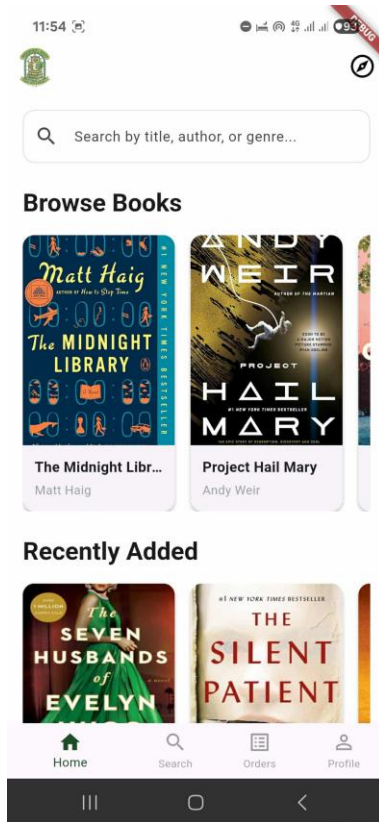
San Juan Campus, San Jose, San Juan, Southern Leyte  
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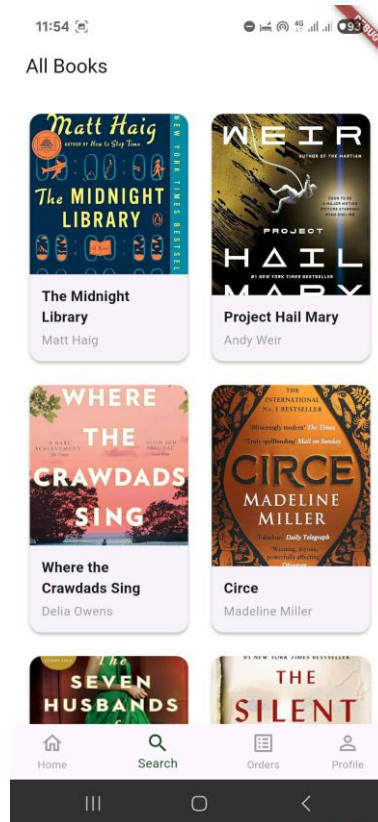
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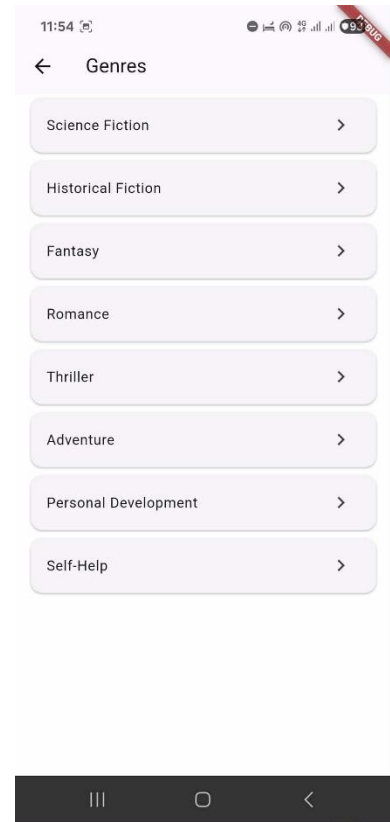
## HOME SCREEN



## SEARCH RESULTS SCREEN



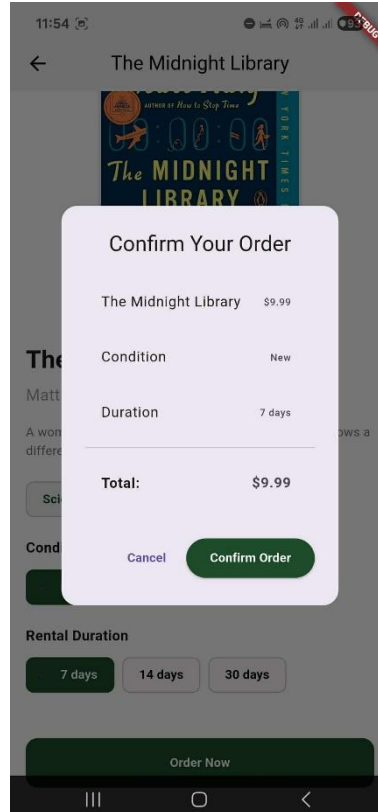
## GENRE SELECTION SCREEN



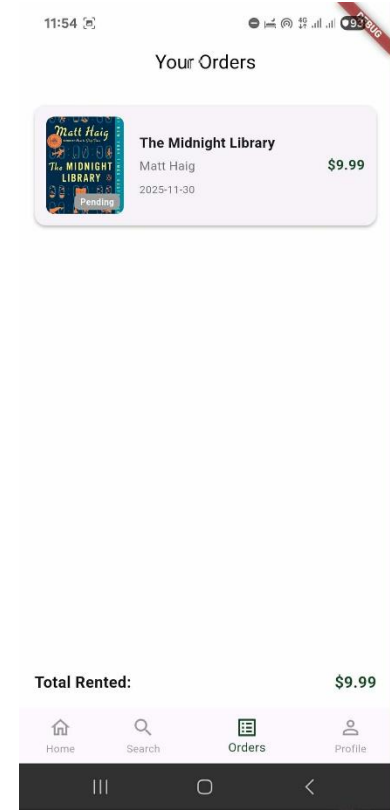
## BOOK DETAILS SCREEN



## CONFIRMATION DIALOG



## ORDERS SCREEN







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## ADMIN LOGIN SCREEN

11:53

**Admin Login**

Enter your email

Enter your password

Login

## ADMIN DASHBOARD SCREEN

11:55

BookRent Admin

**Admin Dashboard**

Total Books: 22

Active Rentals: 0

Pending Orders: 1

Overdue Books: 0

**Recent Activities**

Anamae V. Cubihano pending 'The Midnight Library' 2025-11-30

Dashboard Books Orders

## ADMIN BOOK MANAGEMENT

11:55

Book Management

+ Add New Book

**The Midnight Library** Matt Haig

**Project Hail Mary** Andy Weir

**Where the Crawdads Sing** Delia Owens

**Circe** Madeline Miller

**The Seven Husbands of Evelyn Hugo** Taylor Jenkins Reid

**The Silent Patient** Alex Michaelides

Dashboard Books Orders

## ADD NEW BOOK DIALOG

11:55

Book Management

+ Add New Book

**Add New Book**

Title

Author

Description

Select Genre

No image selected.

Cancel Add

## ADMIN ORDER MANAGEMENT

11:55

Order Management

Order ID: BRK-78048 2025-11-30

**Anamae V. Cubihano**

Pending

\$9.99

Update Status View Details

Dashboard Books Orders

